

AZ-900

Learning path 1: Cloud concepts



Agenda



Learning path 1: Cloud concepts

Learning path 2: Azure architecture and services

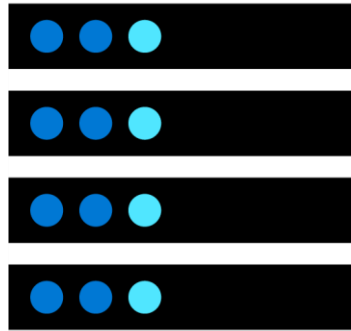
Learning path 3: Azure management and governance

Module 1: Describe cloud computing

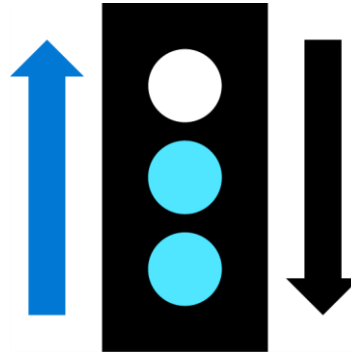


What is cloud computing?

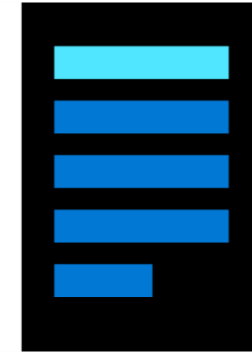
Cloud computing is the delivery of computing services over the internet, enabling faster innovation, flexible resources, and economies of scale.



Compute

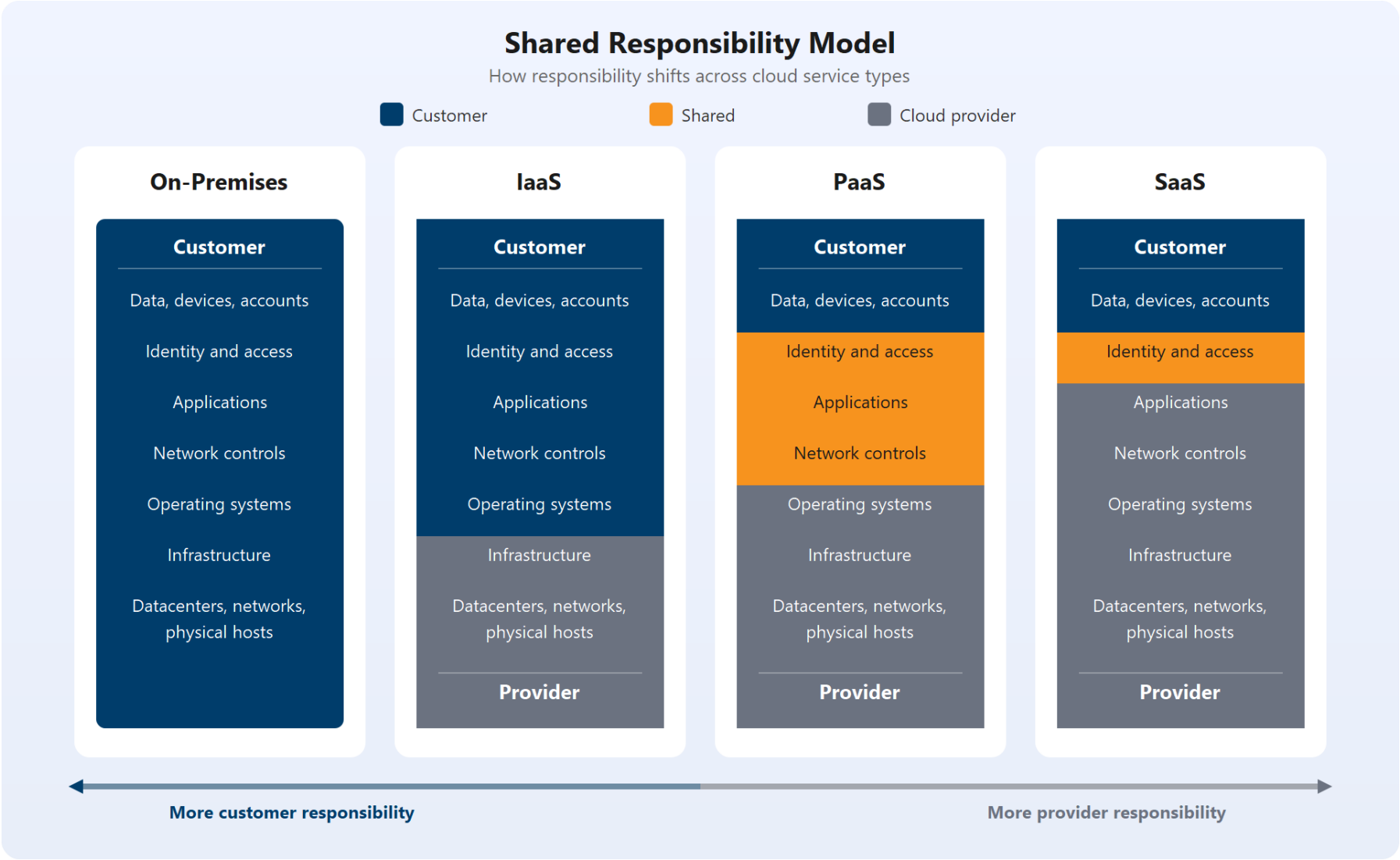


Networking

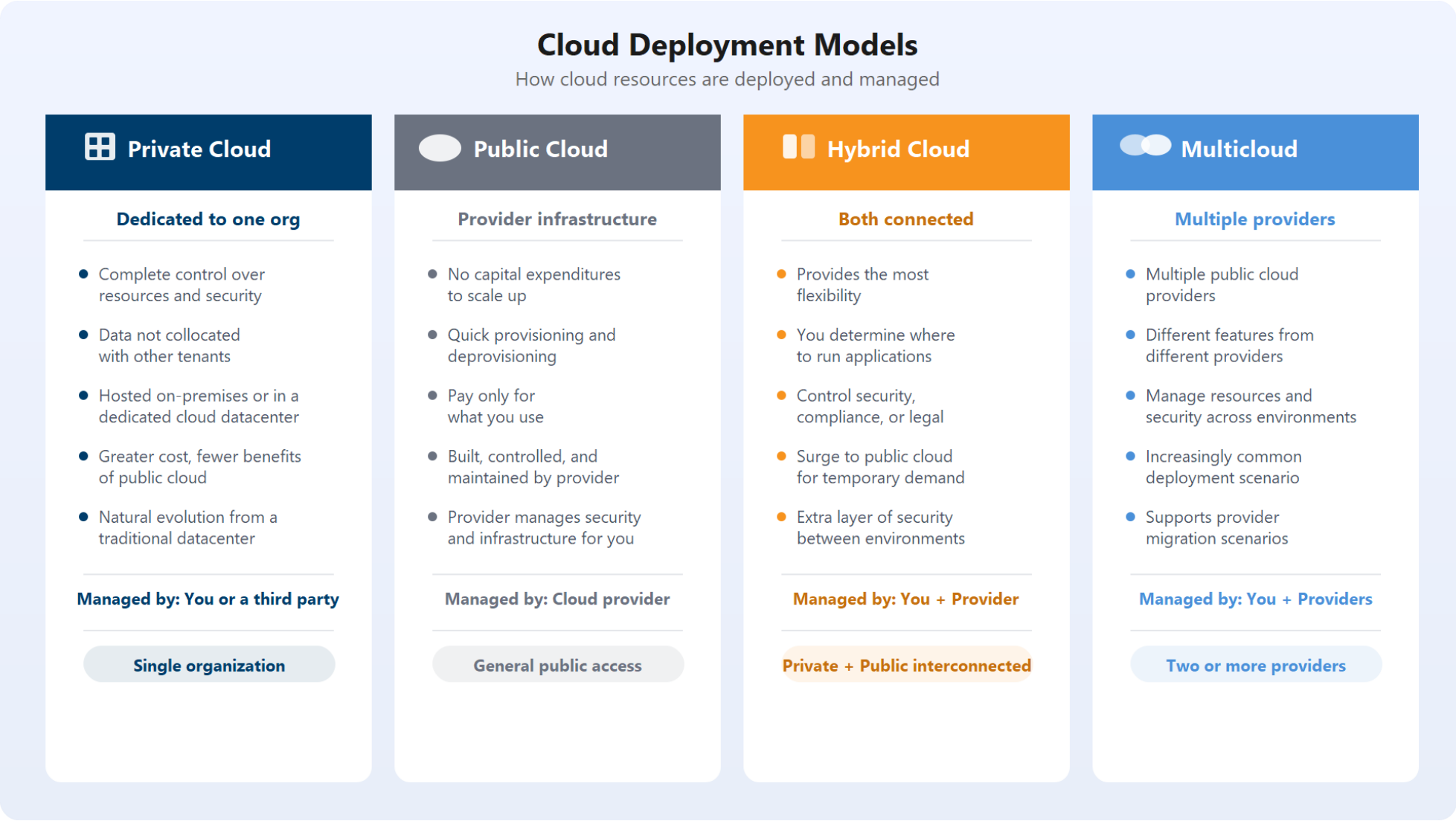


Storage

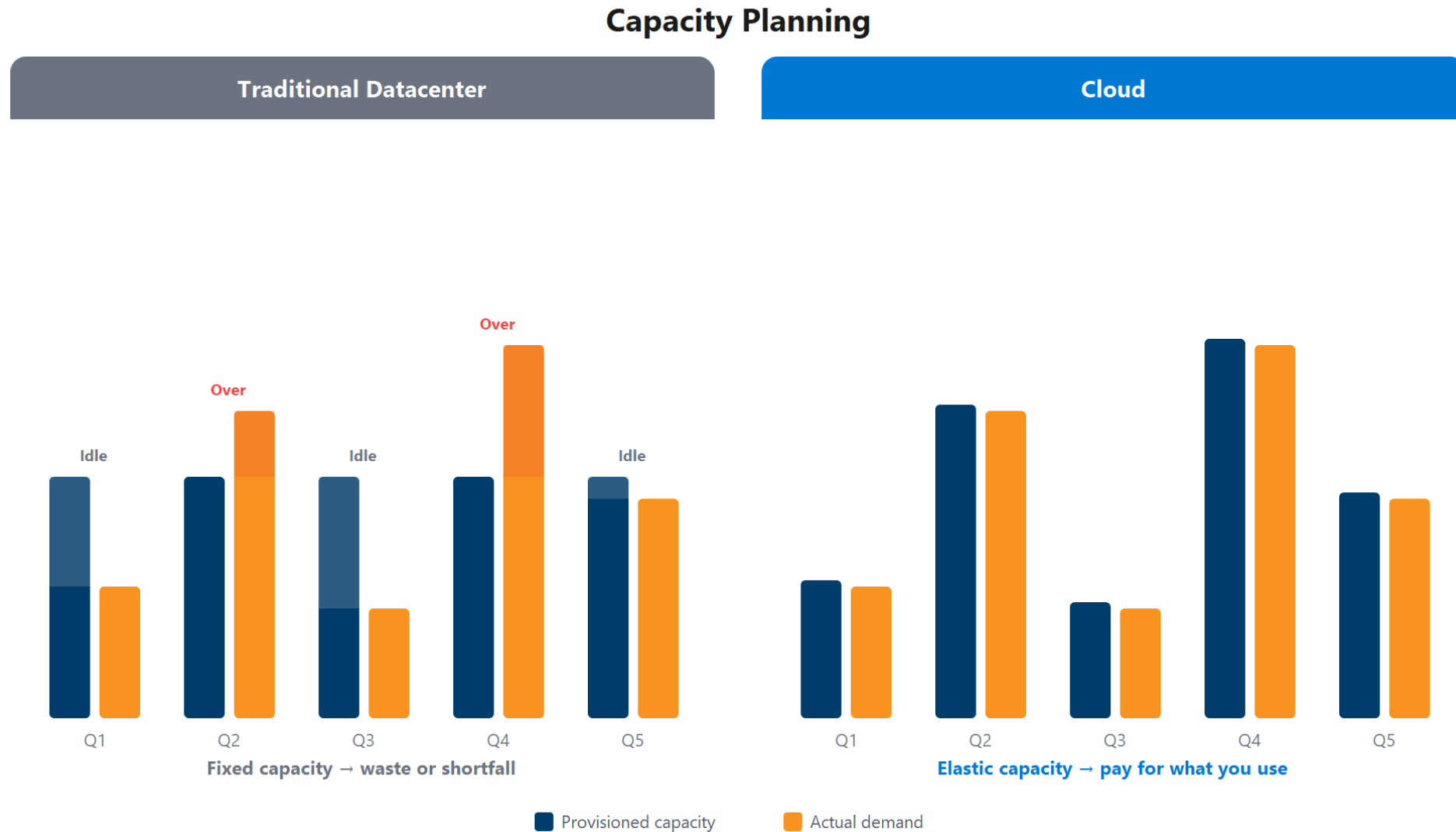
The shared responsibility model



Cloud deployment models



The consumption-based model



Knowledge check



1 In the shared responsibility model, which responsibility always remains with the customer across IaaS, PaaS, and SaaS?

- ☐ Physical hosts
- ☐ Datacenter cooling and power
- ☐ Data and identities

Module 2: Describe the benefits of using cloud services



High availability and scalability

High availability

- Focus on maximum availability regardless of disruptions or events
- Azure SLAs define uptime guarantees for each service
- Architect solutions with high availability in mind

Scalability

- Adjust resources to match changing demand
- Vertical scaling: increase or decrease CPU/RAM of existing resources
- Horizontal scaling: add or subtract the number of resources

Reliability and predictability

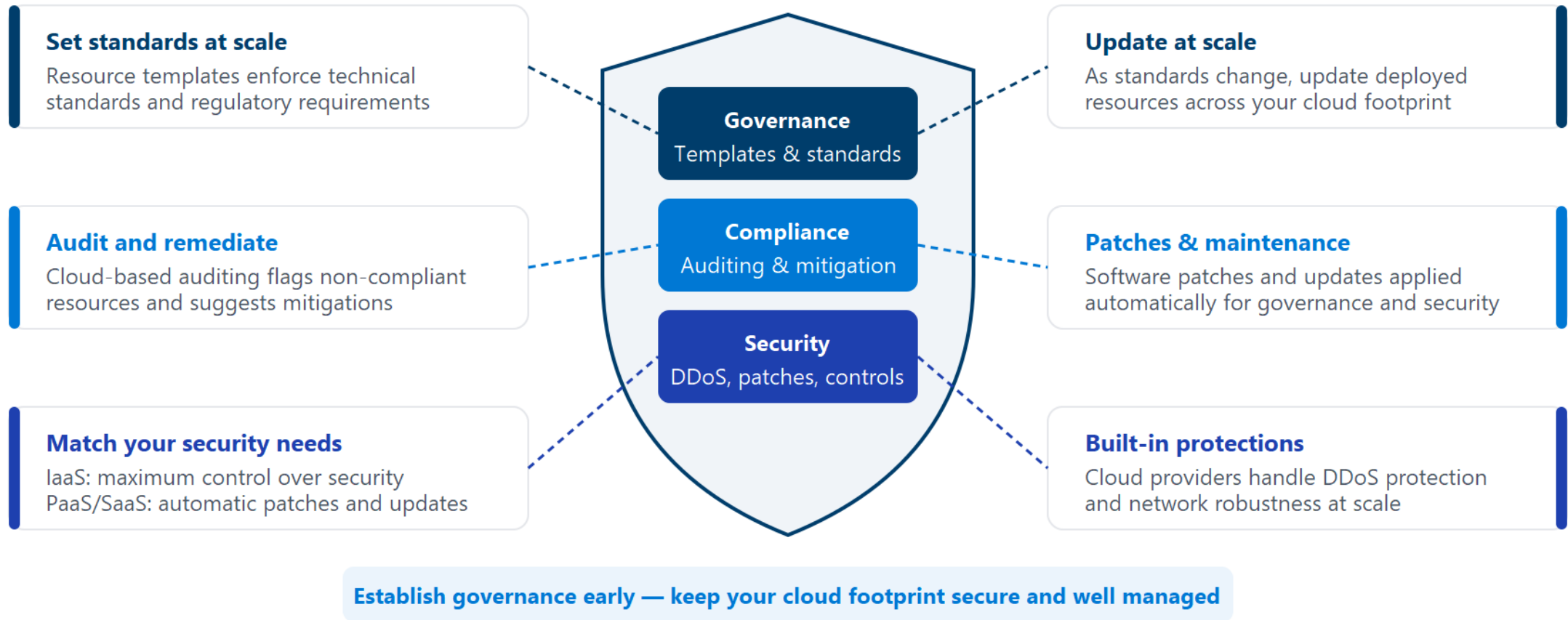
Reliability

- Ability of a system to recover from failures and continue to function
- Part of the Microsoft Azure Well-Architected Framework
- Decentralized design enables deployment in regions around the world
- Automatic failover to a different region in the event of catastrophe

Predictability

- Performance predictability: predict resources needed for a good experience
- Cost predictability: predict or forecast the cost of cloud spend
- Azure Pricing Calculator helps estimate cloud cost

Security and governance in the cloud



Manageability in the cloud

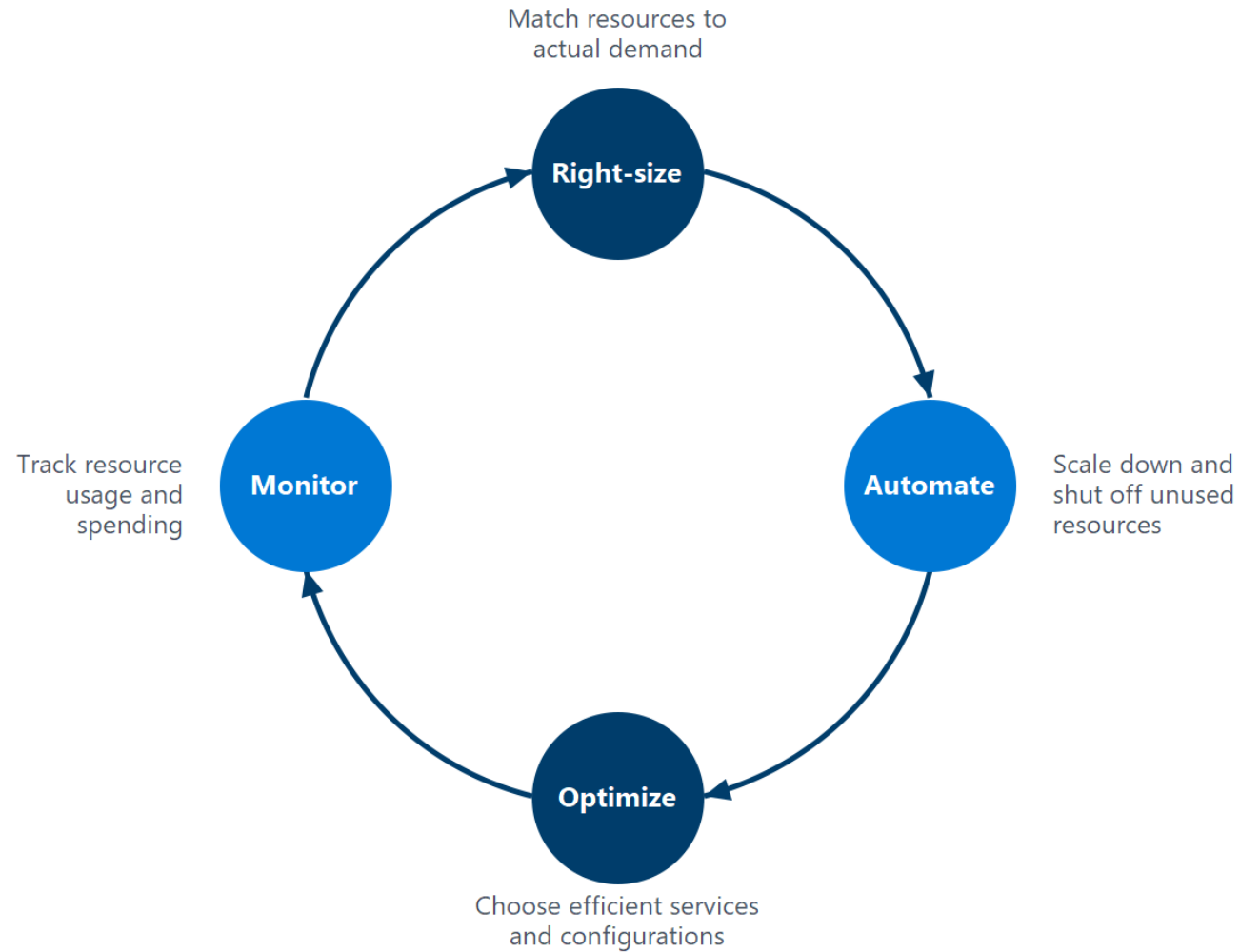
Management of the cloud

- Automatically scale resource deployment based on need
- Deploy resources based on preconfigured templates
- Monitor the health of resources and automatically replace failures
- Receive automatic alerts based on configured metrics

Management in the cloud

- Through a web portal
- Using a command-line interface (CLI)
- Using APIs
- Using PowerShell

Sustainability in the cloud



Right-size, automate, optimize, and monitor — sustainability in action

Knowledge check



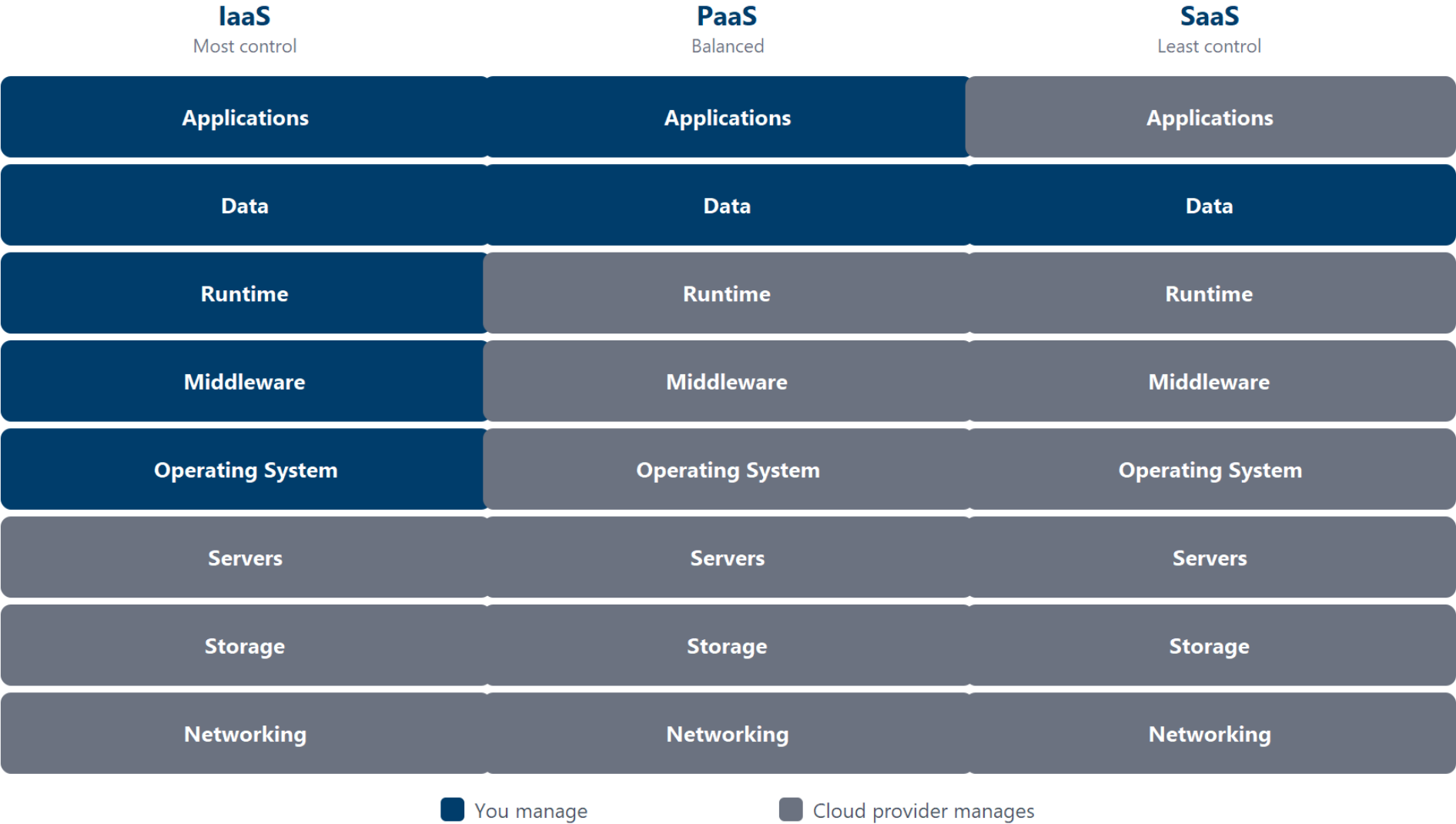
1 Which cloud benefit is best demonstrated by automatically adding resources during peak demand and removing them later?

- ☐ Reliability
- ☐ Elasticity
- ☐ Governance

Module 3: Describe cloud service types

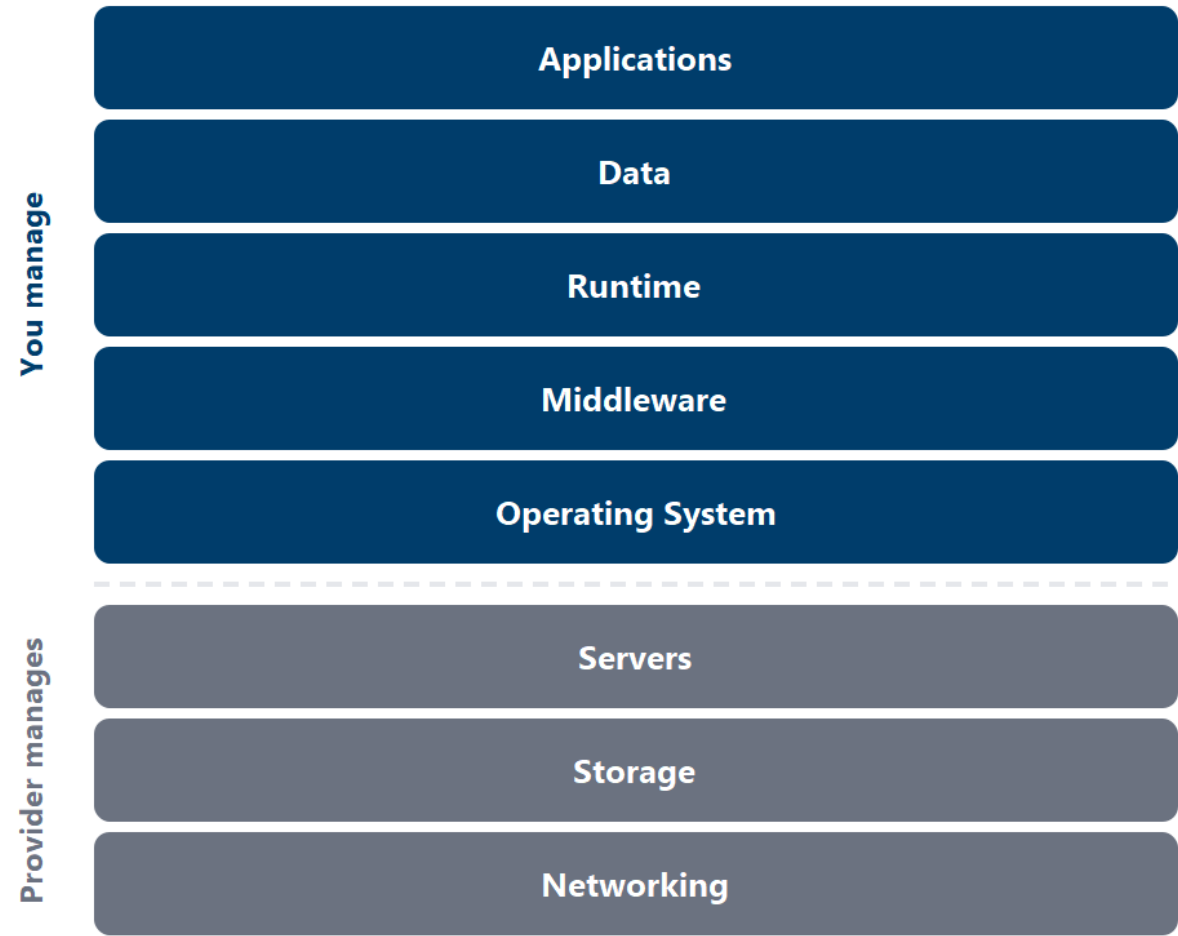


Cloud service types comparison



Infrastructure as a service (IaaS)

Maximum control — you manage the workload stack



Common Scenarios

Lift-and-shift migration

Set up cloud resources similar to on-premises, then move workloads to IaaS infrastructure.

Testing and development

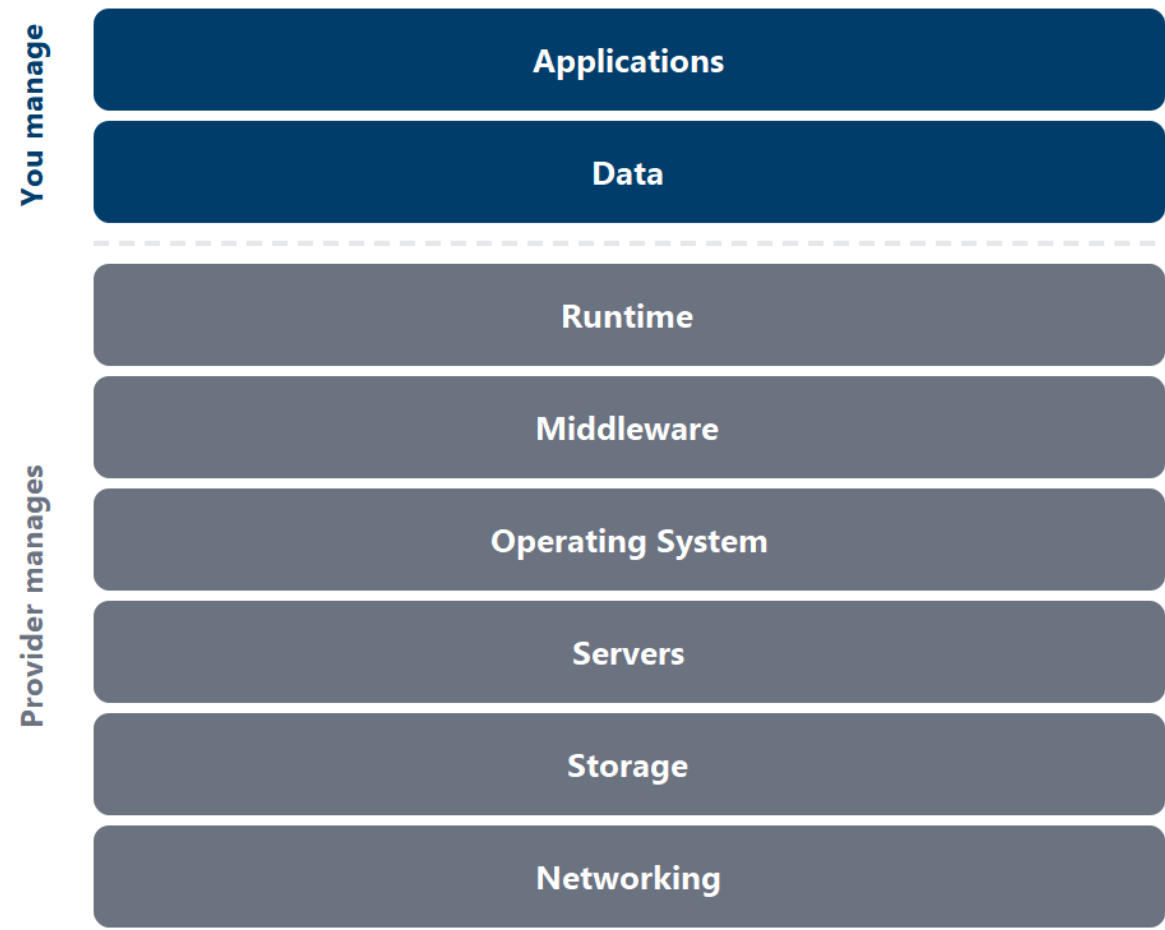
Rapidly replicate dev/test environments. Start up or shut down with complete control.

Key takeaway

You rent hardware — what you do with it is up to you.

Platform as a service (PaaS)

Balanced — you focus on your application and data



Common Scenarios

Development framework

Build cloud-based applications using built-in components with scalability and high availability.

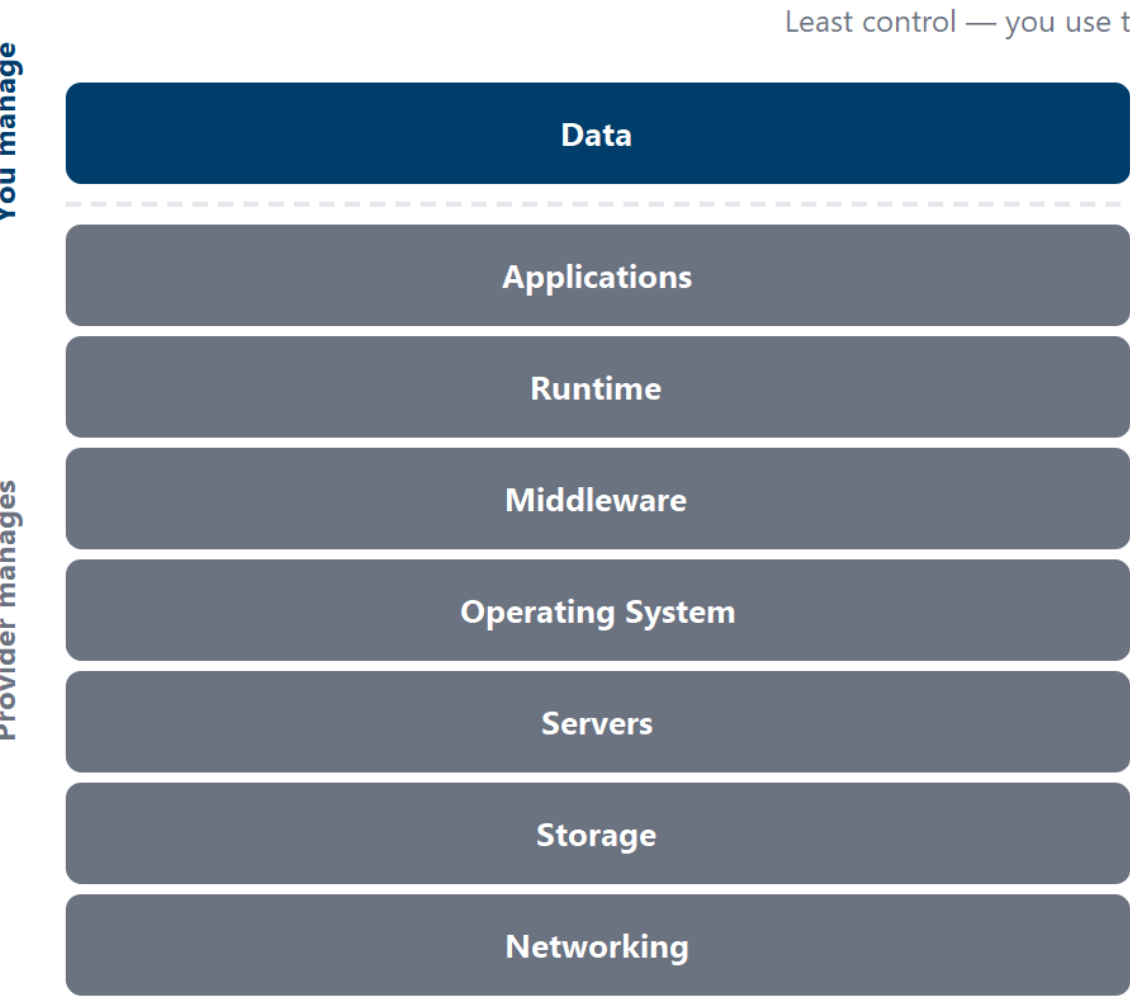
Analytics and business intelligence

Analyze and mine data, find insights, and predict outcomes using platform tools.

Key takeaway

You build on the platform — no infrastructure headaches.

Software as a service (SaaS)



Least control — you use the fully managed application

Common Scenarios

Email and messaging

Microsoft 365, Teams, Outlook

Productivity applications

Word, Excel, PowerPoint online

Finance and expense tracking

Cloud-based financial management tools

Key takeaway

You use the app — the provider handles everything else.

Knowledge check

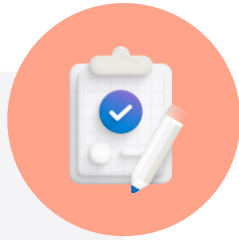


1 Which service type gives the customer the most control over OS configuration and patching?

- ☐ Infrastructure as a Service (IaaS)
- ☐ Platform as a Service (PaaS)
- ☐ Software as a Service (SaaS)

Summary

Learning Path 1: Describe Cloud Concepts



What you learned

- Cloud computing is on-demand delivery of IT services that shifts planning from fixed capacity to flexible consumption.
- Shared responsibility always includes customer ownership of data and identities, regardless of IaaS/PaaS/SaaS.
- IaaS, PaaS, and SaaS differ mainly by how much operational responsibility stays with the customer.

End of presentation



